

Online Quiz System Documentation

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1 Overview

The Online Quiz System is a Java-based desktop application designed to facilitate quiz creation, administration, and participation. It allows administrators to create quizzes with multiple-choice questions, manage quiz data, and view results, while users can take quizzes and view their scores. The application features a user-friendly JavaFX interface and stores data in an SQLite database, ensuring efficient data management and persistence.

Key features include:

- User authentication with admin and user roles.
- Quiz creation with customizable questions and time limits.
- Real-time quiz-taking with a countdown timer.
- Result tracking and display for completed quizzes.

The system is designed for educational environments, providing a robust platform for instructors to assess students and for students to engage in self-paced learning.

2 Running Instructions

To run the Online Quiz System, follow these steps:

1. Prerequisites:

- Java Development Kit (JDK) 22 or later.
- Apache Maven 3.8.0 or later.
- A JavaFX-compatible IDE (e.g., IntelliJ IDEA) or command-line environment.

2. Setup:

- Download the zip file.
- Ensure the project directory contains the `pom.xml` file and the source code in `src/main/java`.

3. Build and Run:

- Open a terminal in the project root directory.
- Run the following Maven command to compile and launch the application:

```
mvn clean compile javafx:run
```

- The application will initialize the SQLite database (`quiz_system.db`) and launch

the login screen.

4. Usage:

- Log in as an admin (e.g., username: admin, password: admin123) to create or manage quizzes.
- Log in as a user to take available quizzes.
- Follow the on-screen instructions to navigate the interface.

Note: Ensure that JavaFX is properly configured in your environment. If issues arise, verify the `pom.xml` dependencies and consult the JavaFX documentation at <https://openjfx.io>.

3 Technologies and Dependencies

The Online Quiz System is built using the following technologies and libraries:

- **Java 22:** Core programming language for application logic.
- **JavaFX 24.0.1:** Framework for building the graphical user interface.
- **SQLite:** Lightweight database for storing user, quiz, and result data.
- **Maven:** Build tool for dependency management and project compilation.
- **JDBC:** Java Database Connectivity for SQLite database interactions.
- **JUnit 5.11.3:** Testing framework for unit tests.

Key dependencies specified in the `pom.xml` file include:

- `org.openjfx:javafx-controls:24.0.1`
- `org.openjfx:javafx-fxml:24.0.1`
- `org.xerial:sqlite-jdbc:3.49.1.0`
- `org.junit.jupiter:junit-jupiter:5.11.3 (test scope)`
- `org.apache.maven.plugins:maven-compiler-plugin:3.13.0`
- `org.openjfx:javafx-maven-plugin:0.0.8`

4 Directory Structure

The project is organized as follows:

- `src/main/java/com/quizsystem/`: Core application code.

- `dao/`: Data Access Objects for database operations.
- `model/`: Data models for users, quizzes, questions, and results.
- `service/`: Business logic for application operations.
- `ui/`: JavaFX controllers and main application.
- `util/`: Utility classes (e.g., database connection).
- `src/main/resources/`: FXML files and CSS stylesheets.
- `pom.xml`: Maven configuration file.

5 Additional Information

The project includes comprehensive Javadoc documentation for all classes, methods, and fields, accessible by running:

```
mvn javadoc:javadoc
```

The generated documentation is located in `target/site/apidocs`.